

Finding an Indicator Function for Palindromic Numbers

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Abstract

In this paper, we present two distinct indicator functions for palindromic numbers. For any positive integer $n \in \mathbb{N}^*$, each function returns 1 if n is palindromic and 0 otherwise. Throughout, integers are written in base 10. The two constructions differ in their approach: the first uses a floor of an arithmetic mean, while the second uses a product over digit pairs.

1 Introduction

Definition 1. Let $n \in \mathbb{N}^*$. Writing n in base 10, there exist a non-negative integer k and digits $a_0, a_1, \dots, a_k \in \llbracket 0, 9 \rrbracket$ with $a_k \neq 0$ such that:

$$n = \sum_{i=0}^k a_i 10^i.$$

The integer n is called a palindromic number if and only if $a_i = a_{k-i}$ for every $i \in \llbracket 0, k \rrbracket$.

We use the notation $\llbracket a, b \rrbracket$ to denote the set of integers $\{a, a+1, \dots, b\}$ for $a \leq b$.

2 Preliminary Propositions

Proposition 1. Let $n \in \mathbb{N}^*$. The number of digits of n written in base 10 is:

$$N(n) = \lfloor \log_{10}(n) + 1 \rfloor.$$

Proof. Writing n in base 10 gives the existence of $k \in \mathbb{N}$ and $a_0, \dots, a_k \in \llbracket 0, 9 \rrbracket$ with $a_k \neq 0$ such that $n = \sum_{i=0}^k a_i 10^i$. The number n has $k+1$ digits. Since $10^k \leq n < 10^{k+1}$, taking $\log_{10}$ yields:

$$k \leq \log_{10}(n) < k+1,$$

so $k+1 \leq \log_{10}(n) + 1 < k+2$, and therefore $\lfloor \log_{10}(n) + 1 \rfloor = k+1$. □

Proposition 2. Let $a, b \in \mathbb{R}_+^*$. Define $f : (\mathbb{R}_+^*)^2 \rightarrow \{0, 1\}$ by:

$$f(a, b) = \left\lfloor \frac{a}{b} \right\rfloor \left\lfloor \frac{b}{a} \right\rfloor = \begin{cases} 1 & \text{if } a = b, \\ 0 & \text{if } a \neq b. \end{cases}$$

Note : We can define f on $\mathbb{R}_+$ by using an $\epsilon > 0$ such as :

$$f(a, b) = \left\lfloor \frac{a + \epsilon}{b + \epsilon} \right\rfloor \left\lfloor \frac{b + \epsilon}{a + \epsilon} \right\rfloor$$

Proof. • If $a = b$, then $f(a, a) = \lfloor 1 \rfloor \lfloor 1 \rfloor = 1$.

- If $a \neq b$, assume without loss of generality $a < b$. Then $0 \leq \frac{a}{b} < 1$, so $\lfloor \frac{a}{b} \rfloor = 0$, and thus $f(a, b) = 0$.

□

Proposition 3. Let $n \in \mathbb{N}^*$. The d -th digit of n in base 10 counting from the right (starting at $d = 1$) is:

$$C_d(n) = \left\lfloor 10 \left(\frac{n}{10^d} - \left\lfloor \frac{n}{10^d} \right\rfloor \right) \right\rfloor.$$

More precisely, if $n = a_k \cdots a_1 a_0$, then $a_i = C_{i+1}(n)$ for every $i \in \llbracket 0, k \rrbracket$.

Proof. Write $n = \sum_{i=0}^k a_i 10^i$. For every $j \in \llbracket 0, k \rrbracket$:

$$\frac{n}{10^{j+1}} - \left\lfloor \frac{n}{10^{j+1}} \right\rfloor = \sum_{i=0}^k a_i 10^{i-j-1} - \left\lfloor \sum_{i=0}^k a_i 10^{i-j-1} \right\rfloor.$$

Since

$$\sum_{i=0}^k a_i 10^{i-j-1} = \underbrace{\sum_{i=0}^j a_i 10^{i-j-1}}_{<1} + \underbrace{\sum_{i=j+1}^k a_i 10^{i-j-1}}_{\in \mathbb{N}},$$

the floor equals the integer part, so:

$$C_{j+1}(n) = \left\lfloor 10 \sum_{i=0}^j a_i 10^{i-j-1} \right\rfloor = \left\lfloor \sum_{i=0}^j a_i 10^{i-j} \right\rfloor = a_j,$$

since $\sum_{i=0}^j a_i 10^{i-j} = a_j + \underbrace{\sum_{i=0}^{j-1} a_i 10^{i-j}}_{<1}$.

□

3 Indicator Functions

Theorem 1. Let $n \in \mathbb{N}^*$ and let $k+1 = N(n)$ be its number of digits. The function $D : \mathbb{N}^* \rightarrow \{0, 1\}$ defined by:

$$D(n) = \left\lfloor \frac{1}{k+1} \sum_{i=0}^k f(C_{i+1}(n), C_{k+1-i}(n)) \right\rfloor$$

satisfies $D(n) = 1$ if n is palindromic and $D(n) = 0$ otherwise.

(This result holds for $k \geq 1$, i.e. $n \geq 10$; single-digit numbers are trivially palindromic.)

Proof. Write $n = \sum_{i=0}^k a_i 10^i$.

Case 1: n is palindromic. For every $i \in \llbracket 0, k \rrbracket$, $a_i = a_{k-i}$, so $C_{i+1}(n) = C_{k+1-i}(n)$, hence $f(C_{i+1}(n), C_{k+1-i}(n)) = 1$. Therefore:

$$D(n) = \left\lfloor \frac{1}{k+1} \sum_{i=0}^k 1 \right\rfloor = \lfloor 1 \rfloor = 1.$$

Case 2: n is not palindromic. There exists $i_0 \in \llbracket 0, k \rrbracket$ such that $a_{i_0} \neq a_{k-i_0}$, so $f(C_{i_0+1}(n), C_{k+1-i_0}(n)) = 0$. Since all terms of f lie in $\{0, 1\}$, the sum is at most k , giving:

$$D(n) = \left\lfloor \frac{1}{k+1} \sum_{i=0}^k f(C_{i_0+1}(n), C_{k+1-i_0}(n)) \right\rfloor \leq \left\lfloor \frac{k}{k+1} \right\rfloor = 0.$$

□

Theorem 2. Let $n \in \mathbb{N}^*$ and let $k+1 = N(n)$. The function $D : \mathbb{N}^* \rightarrow \{0, 1\}$ defined by:

$$D(n) = \prod_{i=0}^k f(C_{i+1}(n), C_{k+1-i}(n))$$

satisfies $D(n) = 1$ if n is palindromic and $D(n) = 0$ otherwise.

Proof. Write $n = \sum_{i=0}^k a_i 10^i$.

Case 1: n is palindromic. Every factor equals 1, so $D(n) = \prod_{i=0}^k 1 = 1$.

Case 2: n is not palindromic. There exists i_0 such that $f(C_{i_0+1}(n), C_{k+1-i_0}(n)) = 0$, so the product contains a zero factor, giving $D(n) = 0$. □

4 Fully Expanded Symbolic Expressions

In this section, we substitute the definitions of f , C_d , and N into the formulas of Theorems 1 and 2 to obtain a single closed-form expression in n alone, with no auxiliary functions.

Recall the three building blocks:

$$N(n) = \lfloor \log_{10}(n) + 1 \rfloor, \quad k+1 = N(n),$$

$$C_d(n) = \left\lfloor 10 \left(\frac{n}{10^d} - \left\lfloor \frac{n}{10^d} \right\rfloor \right) \right\rfloor,$$

$$f(a, b) = \left\lfloor \frac{a + \epsilon}{b + \epsilon} \right\rfloor \left\lfloor \frac{b + \epsilon}{a + \epsilon} \right\rfloor.$$

Using the definition of f we have :

$$f(C_{i+1}(n), C_{N(n)-i}(n)) = \left\lfloor \frac{C_{i+1}(n) + \epsilon}{C_{N(n)-i}(n) + \epsilon} \right\rfloor \left\lfloor \frac{C_{N(n)-i}(n) + \epsilon}{C_{i+1}(n) + \epsilon} \right\rfloor.$$

Substituting the definition of $C_d(n)$ explicitly, this becomes:

$$f(C_{i+1}(n), C_{N(n)-i}(n)) = \left\lfloor \frac{\left\lfloor 10 \left(\frac{n}{10^{i+1}} - \left\lfloor \frac{n}{10^{i+1}} \right\rfloor \right) \right\rfloor + \epsilon}{\left\lfloor 10 \left(\frac{n}{10^{N(n)-i}} - \left\lfloor \frac{n}{10^{N(n)-i}} \right\rfloor \right) \right\rfloor + \epsilon} \right\rfloor \cdot \left\lfloor \frac{\left\lfloor 10 \left(\frac{n}{10^{N(n)-i}} - \left\lfloor \frac{n}{10^{N(n)-i}} \right\rfloor \right) \right\rfloor + \epsilon}{\left\lfloor 10 \left(\frac{n}{10^{i+1}} - \left\lfloor \frac{n}{10^{i+1}} \right\rfloor \right) \right\rfloor + \epsilon} \right\rfloor.$$

Theorem 1 – Fully Expanded

The indicator function of Theorem 1 writes in full as:

$$D_1(n) = \left\lfloor \frac{1}{N(n)} \sum_{i=0}^{N(n)-1} \left\lfloor \frac{\left\lfloor 10 \left(\frac{n}{10^{i+1}} - \left\lfloor \frac{n}{10^{i+1}} \right\rfloor \right) \right\rfloor + \epsilon}{\left\lfloor 10 \left(\frac{n}{10^{N(n)-i}} - \left\lfloor \frac{n}{10^{N(n)-i}} \right\rfloor \right) \right\rfloor + \epsilon} \right\rfloor \cdot \left\lfloor \frac{\left\lfloor 10 \left(\frac{n}{10^{N(n)-i}} - \left\lfloor \frac{n}{10^{N(n)-i}} \right\rfloor \right) \right\rfloor + \epsilon}{\left\lfloor 10 \left(\frac{n}{10^{i+1}} - \left\lfloor \frac{n}{10^{i+1}} \right\rfloor \right) \right\rfloor + \epsilon} \right\rfloor \right\rfloor$$

Theorem 2 – Fully Expanded

Similarly, the indicator function of Theorem 2 writes in full as:

$$D_2(n) = \prod_{i=0}^{N(n)-1} \left[\frac{\left\lfloor 10 \left(\frac{n}{10^{i+1}} - \left\lfloor \frac{n}{10^{i+1}} \right\rfloor \right) \right\rfloor + \epsilon}{\left\lfloor 10 \left(\frac{n}{10^{N(n)-i}} - \left\lfloor \frac{n}{10^{N(n)-i}} \right\rfloor \right) \right\rfloor + \epsilon} \right] \cdot \left[\frac{\left\lfloor 10 \left(\frac{n}{10^{N(n)-i}} - \left\lfloor \frac{n}{10^{N(n)-i}} \right\rfloor \right) \right\rfloor + \epsilon}{\left\lfloor 10 \left(\frac{n}{10^{i+1}} - \left\lfloor \frac{n}{10^{i+1}} \right\rfloor \right) \right\rfloor + \epsilon} \right]$$

These expressions depend solely on n and elementary operations (division, subtraction, floor function). They make explicit that palindrome detection is entirely reducible to integer arithmetic, with no reference to the decimal representation of n as a string.

5 Conclusion

We have presented two closed-form indicator functions for palindromic numbers in base 10. Both rely on the same building blocks – the digit-extraction formula $C_d(n)$ and the equality detector f – but combine them differently: Theorem 1 uses a floored average, while Theorem 2 uses a product. The product formulation (Theorem 2) is arguably simpler and easier to generalize to other bases, whereas the floored-average formulation (Theorem 1) offers an interesting arithmetic structure. Future work could explore extensions to arbitrary bases or to other families of digit-pattern numbers.

References

- [1] G.H. Hardy and E.M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.